

MVP

Multi-Agent Orchestration & Data Infrastructure

Project Kickoff – 8th June 2026

Meet The Team



Abhinav Bharti

Lead Architect



Poojha Subramanian

Project Manager



Amit Vijay Singh

Sr. ML Engineer



**Arul Mozhi Selvan
Ravichandran**

Sr. ML Engineer



MaheshKumar S

DevOps Engineer



Sriya B

ML Engineer



Manikandan J

QA Engineer

Executive Summary



Lofi is an AI-native paid media platform that enables multi-location brands — franchise chains, insurance agencies, and similar businesses — to launch and optimize localized advertising campaigns across 30+ channels in under 60 seconds. At the core of Lofi's product is Lucy, a router-based multi-agent AI system that acts as each user's virtual Head of Paid Media, handling everything from audience building and creative generation to performance analysis and campaign strategy.



Lofi has product-market fit, live enterprise clients including Valvoline, Allstate, and Goosehead Insurance, and a well-documented engineering roadmap. The platform runs on AWS App Runner (Python/FastAPI AI backend), Vercel (Next.js frontend), Supabase (PostgreSQL database), and a provider-agnostic LLM configuration supporting both Anthropic and OpenAI models. Lucy's eight specialist agents currently operate independently — there is no layer to chain them together, pass structured outputs between them, or run multi-step workflows end-to-end.



GoML proposes building this foundation as a focused MVP engagement across five tracks:

- **Multi-Agent Orchestration Engine** — the stateful execution layer that chains Lucy's agents into sequential and parallel, human-check pointed workflows.
- **Campaign Performance Data Pipeline** — automated daily ingestion of ad performance data from Meta, Google, and TikTok into Lofi's pre-designed Supabase metrics tables.
- **Competitive Intelligence Store** — a scheduled, queryable store of competitor ad activity sourced from SearchAPI.io, giving the Market Analyst agent durable historical data to work from.
- **Creative Asset Storage & Metadata** — S3-backed storage with structured metadata linking every generated creative asset to its campaign, variant, and performance outcome.
- **Architecture Advisory & Eval Framework** — fractional AI/ML architecture guidance on database, compute, and model selection decisions, combined with an LLM-as-judge evaluation suite that gives Lofi measurable, reproducible confidence in agent output quality.

Objective

To design and deliver the foundational orchestration, data, and evaluation infrastructure that enables Lofi's Lucy multi-agent system to execute coordinated, end-to-end paid media workflows — unlocking Lofi's five core product workflows and enabling the platform to scale its agentic capabilities with speed, reliability, and measurable output quality.

Scope of Work

1. Multi-Agent Orchestration Engine

Today, a user asking Lucy to “launch a \$50K summer campaign across all 22 locations on Meta and Google” cannot be served end-to-end — no single agent can do it, and there is no layer to coordinate the agents that collectively could. The orchestration engine is the unlock for Lofi's entire product roadmap. Its initial MVP target is Workflow 2: Brief-to-Campaign Launch.

- Design and implement a stateful task graph execution layer that decomposes a high-level user job into a directed graph of agent tasks with explicit dependency resolution
- Support sequential and parallel agent execution within the same workflow — enabling agents like the Market Analyst and Performance Analyst to run simultaneously where their outputs are independent
- Define typed Pydantic v2 I/O schemas for every inter-agent handoff (e.g., CreativeResearchOutput → CreativeBrief → CreativeAssetSet → QAReport) — eliminating ambiguity and fragility between agents
- Implement human-in-the-loop checkpoint support — pause workflow at defined gates, surface intermediate output to user for review, resume only on explicit approval
- Handle revision loops — when a QA step fails, route back to the specific producing agent rather than restarting the full pipeline
- Implement error recovery and partial completion handling — preserve state of all completed steps on failure, surface actionable failure states to user and operator
- Extend the existing /stream NDJSON protocol with workflow-level event types for real-time frontend progress rendering (agent running, complete, awaiting review, errored)
- Persist workflow execution state — task graph, per-step status, intermediate outputs, checkpoint decisions — in Supabase, queryable by both agents and the frontend

2. Data Infrastructure — Foundation Layer

2A. Campaign Performance Data Pipeline

Lofi has pre-designed campaign_metrics and audience_metrics tables in Supabase but no automated ingestion pipeline. The Performance Analyst agent and any orchestrated workflow that depends on performance context cannot function at scale without fresh, reliable data in these tables.

- Build API connectors for Meta Ads, Google Ads, and TikTok Ads to pull campaign and audience performance data at daily grain — by location, ad variant, and demographic dimension
- Implement data normalization logic to map each platform's native schemas and metric names into Lofi's unified table schemas, storing both raw and computed metrics (ROAS, CPA, CPC, CTR) at ingest time
- Build an incremental daily refresh pipeline with support for retroactive corrections when platforms revise reported figures
- Implement error handling for API failures, rate limits, and schema changes — with logging, alerting, and configurable retry logic
- Ensure pipeline idempotency: re-running a day's pipeline produces the same result without duplicate rows
- Validate data integrity post-ingestion: row counts, metric range checks, and cross-platform consistency

2B. Competitive Intelligence Store (Initial Version)

Today the Market Analyst agent performs one-shot competitive analytics but there is no durable, scheduled, or queryable store of competitor ad data. Competitive trends cannot be tracked over time, and no agent can draw on historical competitor intelligence without re-fetching it live.

- Design a structured Supabase schema to store competitor ad data - capturing creative content, ad copy, platform, date range, spend signals, and targeting signals per competitor
- Build an ingestion pipeline with configurable monitoring schedules per competitor
- Implement deduplication and retention policies to prevent record proliferation across repeated ingestion runs
- Ensure the store is queryable by the Market Analyst agent via SQL tool calls, with structured lookups by competitor, time window, and platform
- Deliver an initial competitor tracking configuration for reference competitors as part of the MVP

2C. Creative Asset Storage & Metadata (Foundation)

Lucy's Image Designer and Video Designer agents generate creative assets but these are not systematically stored, versioned, or linked to the campaigns, A/B test variants, or performance outcomes they belong to. Performance-driven creative iteration is impossible without this foundation.

- Provision S3 asset storage with bucket configuration, lifecycle policies, and IAM roles consistent with Lofi's existing AWS Terraform infrastructure
- Design and implement a Supabase metadata schema linking stored assets to: campaign ID, ad variant, A/B test hypothesis, producing agent run, creative brief ID, platform, and creation date
- Implement asset version history to support iterative creative refinement — each new version references its predecessor
- Ensure the schema supports future linkage of assets to performance outcomes in campaign_metrics
- Integrate asset storage and metadata writes into the orchestration engine so that assets produced within a workflow are automatically stored and registered without manual steps

3. Architecture Advisory & Optimization

Lofi's engineering team has open questions on Supabase's long-term fit under multi-agent load, model selection per agent role, and AWS compute sizing as Lucy's footprint grows. GoML will contribute a fractional AI/ML architect throughout the engagement to provide ongoing guidance on these decisions

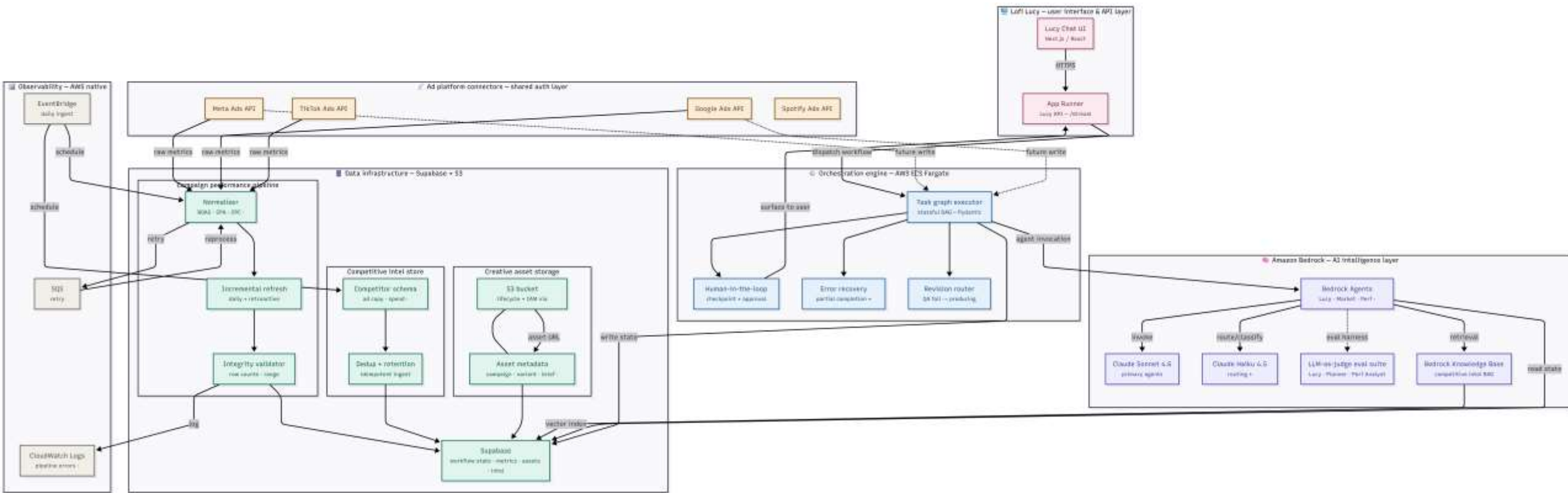
- Assess Lofi's Supabase architecture against projected query patterns under multi-agent orchestration load and deliver a written recommendation on retaining, migrating, or adopting a hybrid database approach
- Provide guidance on evolving the current pydantic-ai orchestrator-worker pattern toward a production-grade stateful task graph execution model
- Advise on AWS compute architecture for multi-agent workloads: App Runner instance sizing, auto-scaling configuration, and cost implications of 5–8 agent invocations per workflow
- Review and validate LLM model tier configuration per agent role (claude-sonnet-4-6 for primary agents, claude-haiku-4-5 for routing and classification) in the context of orchestrated multi-step workflows
- Design and implement an LLM-as-judge eval suite covering at least three Lucy agents (Lucy general, Campaign Planner, Performance Analyst) with calibrated judge prompts, a reproducible test harness, and a regression-detection scoring report

4. Integration Readiness (Scoped to MVP Needs)

The ad platform connectors built for the Campaign Performance Data Pipeline also form the read-side foundation for the future Campaign Builder agent's writeconnectors. Integration readiness work in this phase ensures the architecture is designed once and extended cleanly in the next phase.

- Implement all ad platform connectors (Meta, Google Ads, TikTok, Spotify) with clean separation between read operations (performance ingestion) and writeoperations (future campaign launch), using a shared authentication and session management layer
- Validate API authentication flows, permission scopes, and rate limit handling across all three platforms for both daily batch ingestion and potential near-real-time use
- Document connector interfaces and data models so that write-side campaign builder connectors can be implemented in a future phase without redesigning theauthentication or normalization layers
- Review the existing Supabase schema for campaigns, locations, platforms, and ad variants against connector outputs — flag any schema gaps before theybecome downstream blockers

Architecture Diagram



Proposed Tech Stack



Input, Intel (Processing Layer), Output

Input Layer

- **User jobs** entered via the Lofi chat interface (e.g., "Launch a Q3 campaign for all 22 locations, \$200K, Meta and Google")
- **Brand context** from Supabase — identity, tone of voice, audiences, locations, creative assets, competitor list, keyword blacklist
- **Campaign performance data** from campaign_metrics and audience_metrics (populated by the new pipeline)
- **Competitor ad data** from the Competitive Intelligence Store (Supabase, sourced via SearchAPI.io)
- **Ad platform data** from Meta, Google Ads, and TikTok APIs via new connectors
- **Creative assets** from S3 with Supabase metadata (version history, campaign linkage)
- **Conversation history** from lacy_messages
- **File attachments** processed by the existing file_analysis_tool

Orchestration Layer

- **Task graph engine** — decomposes user job into agent steps with dependency resolution
- **Sequential and parallel execution manager**
- **Pydantic v2 inter-agent data contracts** — typed I/O schemas between every agent handoff
- **Workflow state store** — task graph, per-step status, intermediate outputs, checkpoint decisions in Supabase
- **Human-in-the-loop checkpoint handler** — pause, surface, resume on approval
- **Revision loop router** — failed QA routes to producing agent, not full pipeline restart
- **Extended NDJSON event stream** — real-time workflow progress to frontend

Input, Intel (Processing Layer), Output

Agent Execution Layer

- **Router Agent** — classifies top-level intent, triggers workflow or single-agent response
- **Market Analyst, Performance Analyst** — parallel where workflow dependencies allow
- **Campaign Planner, Creative Director** (including internal sub-agent orchestrator), **Image Designer, Video Designer, Lucy general, Support Agent**
- **All agents** receive the full ChatDeps context: brand context, conversation history, performance data, creative assets, web search

Output Layer

- **Workflow progress events** streamed to the Lofi frontend in real-time
- **Intermediate outputs** surfaced at human-in-the-loop checkpoints for user review before production continues
- **Final workflow outputs** — campaign schemas, creative asset sets, performance reports, media plans — persisted to Supabase and surfaced in the Lofi UI
- **Creative assets** automatically stored to S3 with metadata written to Supabase at workflow completion
- **Competitive intelligence records** written to the Competitive Intelligence Store on scheduled cadence
- **Eval scores and benchmarking reports** for Lofi's internal engineering use

Prerequisites

- Access to the detailed Lofi engineering brief (shared post-NDA by Malte and Cosimo)
- Supabase project credentials — read access for schema review, write access for pipeline and orchestration development
- Meta, Google Ads, Spotify and TikTok developer app credentials and API access
- API credentials for competitive intelligence ingestion
- AWS App Runner and ECR access for deployment alignment
- Access to the lucy-app GitHub repository for orchestration integration
- Sample of historical campaign performance data for pipeline validation and eval calibration
- SME availability from Lofi's engineering team for weekly syncs, schema reviews, and checkpoint approvals during build

Success Criteria

- Orchestration engine executes Workflow 2 (Brief-to-Campaign Launch) end-to-end in staging with parallel agent execution, human-in-the-loop checkpoint functioning correctly, and revision loop routing confirmed on a simulated QA failure
- Workflow state persisted in Supabase and queryable by both the orchestration layer and the frontend event stream
- Campaign performance pipeline ingests daily data from Meta and Google into campaign_metrics with less than 2% data error rate, with idempotent runs confirmed
- Competitive intelligence pipeline populates the schema for at least one configured competitor on a scheduled cadence with deduplication functioning
- Creative assets produced within a workflow automatically stored to S3 and registered in Supabase with correct campaign, variant, and agent-run linkage
- Architecture advisory delivers written database and compute recommendation within the first three weeks, before pipeline or orchestration development locks in design decisions
- Eval framework produces reproducible quality scores for at least three Lucy agents across identical input sets, with a CI-compatible run mode documented

Deliverables

- Multi-agent orchestration engine: task graph executor, parallel/sequential runner, Pydantic v2 I/O schemas, checkpoint handler, revision loop router, errorrecovery, workflow state persistence
- Extended NDJSON streaming protocol with workflow-level event types and frontend integration documentation
- Ad platform API connectors: Meta, Google Ads, TikTok (read, write-extensible architecture)
- Campaign performance data pipeline: normalization layer, incremental refresh, backfill support, idempotency, error handling and alerting
- Competitive Intelligence Store: Supabase schema,, deduplication, retention policy, Market Analyst agent SQL query integration
- Creative Asset Storage: S3 provisioning (Terraform), Supabase metadata schema with version history, orchestration engine integration for automatic assetregistration
- Architecture advisory: written database and compute architecture recommendation, orchestration pattern guidance, model selection review
- LLM-as-judge eval suite: judge prompts, test harness, calibration, benchmarking report for three Lucy agents
- Integration readiness documentation: connector interface specs, authentication layer design, write-side extensibility notes, Supabase schema gap analysis
- Comprehensive technical documentation: architecture overview, pipeline runbooks, orchestration design guide, API connector guides

Out of Scope

- Frontend UI development for the mission control workflow view — GoML delivers the backend event stream and state store; the UI is Lofi's internalresponsibility
- Ad platform write connectors (campaign creation, modification, bid and budget adjustment) — Layer 3, future phase; integration readiness in this MVPensures architecture is extensible for it
- CRM integrations (HubSpot, Salesforce) — lower priority than ad platform connectors for this phase
- New agent builds not yet in Lucy's roster: Growth Planner, Copywriter, Platform Policy Specialist — future phase additions required for Workflows 1 and 3in full
- LLM model fine-tuning or custom model training
- Voice or phone channel integrations, Multi-language support for agent outputs and Historical data migration or cleanup of existing Supabase tables

Limitations

- Multi-agent orchestration significantly multiplies compute and LLM API costs — a single workflow may invoke 5–8 agents, each with multiple LLM calls; AWScompute and API credit coverage should be confirmed with Jack and the Lofi AWS manager before engagement begins
- Data pipeline coverage limited in this phase to Meta, Google Ads, and TikTok; additional platforms can be added in future phases using the shared connector architecture
- Competitive intelligence ingestion scoped to only one data source in this phase
- Eval framework calibration quality depends on availability of human-rated agent output examples during onboarding; judge accuracy improves with additional calibration data over time
- Supabase retained as the primary database for this phase; if architecture advisory recommends migration, the migration itself is a future phase engagement

Future Scope

- Layer 3: Ad platform write connectors — campaign creation, modification, budget and bid adjustment, status sync (Meta, Google, TikTok); Campaign Builderagent goes live
- Workflow 1 (Strategy & Planning), Workflow 4 (Performance Review & Optimization), Workflow 3 (Research-Driven Creative Production), and Workflow 5(Competitive Response) full orchestration
- New agent builds: Growth Planner, Copywriter, Platform Policy Specialist — completing Lucy's full five-workflow roster
- Cross-agent persistent memory — agents access intermediate outputs from other agents' prior runs beyond shared chat history
- Database migration — structured migration of Supabase components to RDS, Aurora, or hybrid architecture if recommended by advisory
- Creative-to-performance linkage — systematic connection of generated creative assets to downstream performance outcomes in campaign_metrics,enabling performance-driven iteration

Proposed Timelines

All prerequisites, data, and infrastructure (servers, tools, access) must be provided and set up by the end of Week 0 to ensure project readiness*

Tasks & Timelines	Week 0				
Gathering Prerequisites					
Infrastructure Access					
Data and Logic Understanding Session					

The project will only begin once all prerequisites are received, the data is understood, and infrastructure is fully operational*
SME availability is required from the client side to validate the test results post deployment*

Tasks	Week 1 (Jun 8 – Jun 12)				Week 2 (Jun 15-19)				Week 3 (Jun 22-26)				Week 4 (Jun 29-Jul 3)				Week 5 (Jul 6-10)				Week 6 (Jul 13-17)				Week 7 (Jul 20-24)				Week 8 (Jul 27-31)				Week 9 (Aug 3-7)				Week 10 (Aug 10-14)			
Project kickoff & AWS Set up																																								
Orchestration engine design																																								
Workflow schema & task graph implementation																																								
Streaming protocol enhancement																																								
Campaign data pipeline development																																								
Competitive intelligence ingestion setup																																								
Creative asset storage implementation																																								
Eval framework design & deployment																																								
Integration of agents with orchestration layer																																								
Integration testing & QA																																								
UAT & refinement																																								
Documentation & Knowledge Transfer																																								

Governance



Executive Sponsors

Assessment of overall health of Engagement

- Engagement vision and discuss future roadmap
- Review strategic objectives in progress
- Brainstorm new ideas for product engagement

- Jack Singer
- Kapil Joshi
- Lofi team

Overall Performance, Strategic Planning

Project Completion

Project Progress Review

Progress Review, Contractual issues

- Delivery Status Reports
- Timelines discussion
- Issues & escalations
- Risks & mitigation

- Poojha S
- Abhinav Bharti
- Lofi team

Delivery Timelines, Escalations, Risks

Weekly

Daily Stand Up

Delivery Management

- Task Management
- Dependency highlights & delays
- Issues resolution
- Team Coordination

- Poojha S
- Abhinav Bharti
- Amit Vijay Singh
- Arul Mozhi
- Manikandan
- Maheshkumar
- Sriya B
- Lofi team

Backlogs, Sprint discussion, Task Management

Daily/Alternate Days/Twice a Week

Escalation Management

Strategic Direction

THANK YOU

Rishabh Sood

Chief Business Officer

Rishabh.sood@goml.io

+91-9582394818

marketing@goml.io